From Desktop Telemetry to Automation Opportunity

Evidence-based process selection and domain research

Dataset B process analysis, candidate selection, and the automation landscape for the selected opportunity.

This document records how a single automation opportunity was selected from unlabelled desktop-operation telemetry, and what the subsequent domain research established about how that class of work is actually automated in practice.

Its purpose is traceability. Every recommendation here can be followed back through the candidate comparison, to the Dataset B metrics, to the Step 1 execution inference, and forward into the requirements that the Step 3 system must satisfy. It is deliberately not a design document: it establishes *what properties* the automation must have, and why.

ASSIGNMENT

From operation logs to an automation proposal. Step 2 asks which processes should be automated, in priority order, with the reasoning behind that ordering.

AUTHORITATIVE SOURCES

`build/dataset_b/step2/process_inventory.json`
`build/dataset_b/step2/gates.json`
`build/dataset_b/segments_rich.jsonl`

STATUS

Step 1 frozen. Step 2 complete. Step 3 not started.

EVIDENCE TAGS

DATA	Measured in Dataset B by our own analysis.	VENDOR	Vendor primary documentation.
LIT	Peer-reviewed or scholarly literature.	OSS	Open-source project.
JP	Japanese government, corporate or industry source.	INF	Our inference.

These are never blurred. Vendor documentation is not treated as empirical evidence, and no production capability is presented as verified.

CONTENTS

1 Executive summary 3
2 Problem and objective 4
3 From telemetry to business processes 5
4 The Dataset B process landscape 6
5 Determinism and transfer: the suitability evidence 7
6 Screening: three gates, declared before the numbers 9
7 Impact × suitability 9
8 The shortlist 10
9 The multi-system discovery 11
10 The document-type discovery 12

11 Alternatives considered, and why they were not selected 13
12 Reframing the opportunity: 二重入力 14
13 The automation modality landscape 15
14 The critical question: is there an import or API path? 16
15 Multi-system consistency and partial failure 17
16 Human-in-the-loop 19
17 AI and LLM agents: assessed, and excluded from the execution path 20
18 Process definitions and reusability 21
19 Design requirements derived from the research 22
20 The evidence boundary 23
21 Decision, priority order, and what would overturn it 24
22 What Stage 3 must decide 25
References 26

HOW TO READ THIS DOCUMENT

IF YOU HAVE FIVE MINUTES	§1 states what was selected and why, and §21 states the priority order and what would overturn it. Everything between them is the evidence for those two pages.
IF YOU INTEND TO DISAGREE	Start at §20. It separates what was measured from what was inferred from what we refused to estimate, and lists the five claims this document deliberately does not make.
IF YOU ARE BUILDING STEP 3	§19 is the requirement list, each item carrying the evidence that produced it, and §22 is the set of decisions this research leaves open.
IF YOU WANT THE NUMBERS	§4 and §5 carry every per-process value used anywhere in this document; both are generated from the frozen Step 1 output.

Figures are numbered continuously and referenced by number throughout; every quantitative claim carries an evidence tag.

1 Executive summary

We were given 15 sessions of unlabelled desktop-operation telemetry from a Japanese company's back office and asked where automation would have the greatest impact. Step 1 reconstructed **183 business-process executions** across **13 process labels** and **11 route families**, covering **7,642.7 seconds** of segmented activity performed by **4 operators**. DATA

This document records what happened next: how those executions were characterised, how candidates were compared, why one was selected, and what the domain research then established about how that class of work is actually automated.

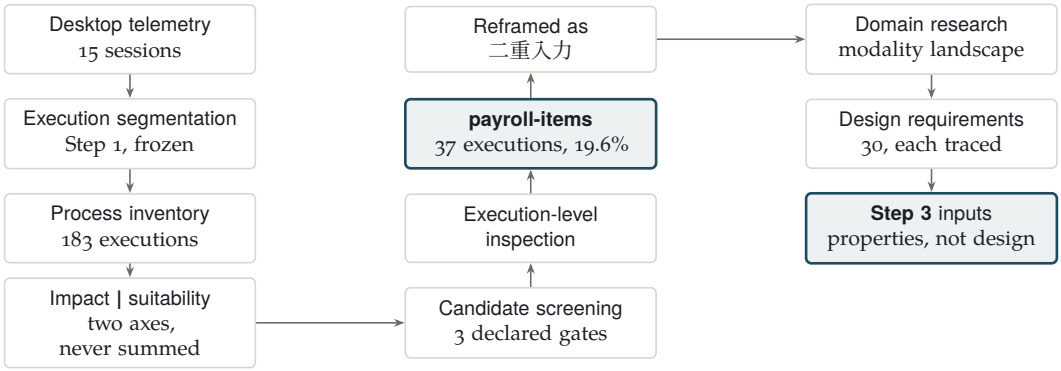


Figure 1. The decision chain. Each step is auditable against a repository artifact; the turn at *execution-level inspection* is where the interpretation of the opportunity changed.

What the analysis found

Frequency and automatability point in different directions. DATA The largest process by observed time — `resident-tax`, 19.2% — is among the *least* repeatable things in the dataset: route-set Jaccard **0.345** (only `social-insurance__p04` is lower, at 0.335), surface-set Jaccard **0.130** — the lowest outright — and no two of its 23 executions share an identical surface shape. Six labels band High on automation suitability and none of them bands High on impact. No process in Dataset B is High on both. Any recommendation is therefore a trade, and this document makes it explicitly rather than hiding it inside a composite score.

The selected opportunity is the `payroll-items` family — 37 executions, 19.6% of segmented time, route-set Jaccard 0.74, route concentration 1.00, 39 distinct field keys. DATA

The critical discovery came from looking inside executions, not at aggregates. DATA The route `/payroll-items` exists in *three distinct enterprise systems* — HR人事給与システム (HR and payroll), 財務会計システム (financial accounting) and 受発注在庫管理システム (order and inventory management) — and it is the only module with substantial activity in all three. The companion document is an 一覧 (a *list*: data), where the companion documents of the rejected candidates are 手続き (*procedures*: reference material a person consults). The opportunity is therefore not document-to-form transcription. It is **duplicate master-data entry across siloed core systems**, with the operator acting as the integration layer.

The domain research then changed the recommended solution shape. JP Japanese integration practice names this problem 二重入力 and its documented consequence 転記ミス — transcription error — and records that CSV import is a *common* route between payroll and accounting packages. Whether the client's systems offer such a route is unverified, and observing staff use a browser is not evidence that no other interface exists. INF UI automation is therefore a *conditional* implementation path, and the system's target adapter must be swappable rather than hard-wired.

AI was assessed and excluded from the execution path, with a number. LIT On WorkArena — enterprise form filling and data entry on a real platform — GPT-4 reaches **42.7%** task success and the open-source state of the art **17.9%**. Against a deterministic path driving element identifiers recovered from the client's own telemetry, an agent would be worse on success rate, reproducibility, explainability and auditability, and its failures would be silent rather than visible.

What remains unresolved is recorded as validation questions rather than filled with assumptions: whether an import or API path exists in each system and what fields it covers; real selector stability; true volumes; error and rework rates; business criticality; and who would own and maintain the automation.

2 Problem and objective

The client's request was stated in business terms: *use these logs to tell us where automation would have the greatest impact on our operations, and show us something that actually works.* Nothing in the logs says which business process any event belongs to.

2.1 Why frequency is not the answer

The intuitive reading — automate whatever consumes the most time — is contradicted by both the external evidence and our own data.

LIT Two independent systematic reviews report that choosing the wrong process is the leading documented cause of automation project failure [6, 3]. JP Japan's Ministry of Internal Affairs and Communications publishes selection criteria for local-government RPA in which *four of five* criteria are properties of the work — routine, requiring no complex judgement; an established *electronic* input source; a *stable* processing target — and only one is volume [10]. VENDOR UiPath's own candidate-vetting score is an average of five factors, none of which is volume [14].

DATA And in Dataset B the two properties are close to inversely related among the largest processes, which is shown directly in Fig. 6.

2.2 Two axes, deliberately not summed

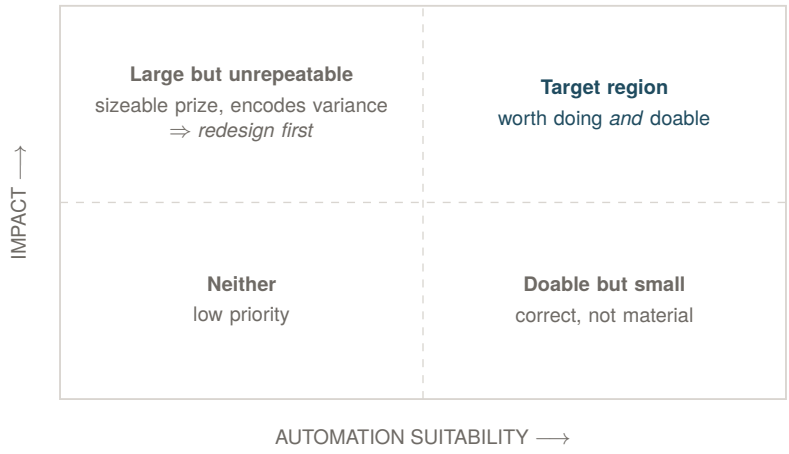


Figure 2. The conceptual frame. Impact answers *is this worth doing*; suitability answers *can this be done reliably*. They are different questions with different evidence, and collapsing them into one number would hide which side carried the decision.

INF The reason for keeping them apart is not stylistic. Every multi-criteria method in the literature — Delphi weighting, AHP/TOPSIS [5], the quantifiable selection method of Wanner et al. [4] —

requires weights elicited from process owners. We had no access to the client's experts. Inventing weights would have produced a number whose precision we could not defend, on a sample of 183 executions from a single capture window.

3 From telemetry to business processes

This section exists for traceability, not to repeat the Step 1 report.

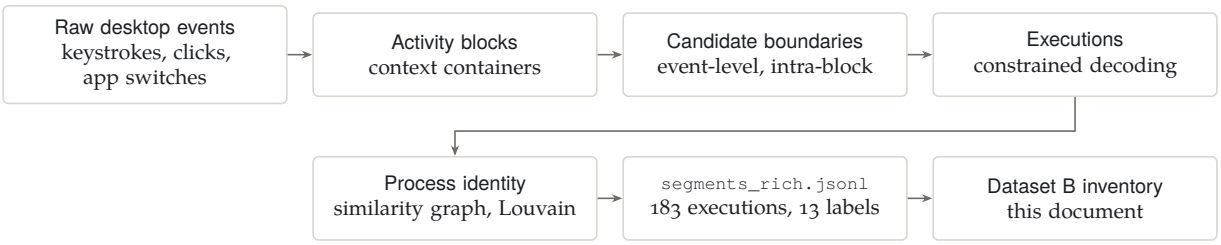


Figure 3. The Step 1 pipeline, frozen, and where Step 2 attaches to it. Step 1's measured quality on the ground-truthed Dataset A was boundary F1 **0.690** at $\tau = 2\,\text{s}$ against a reachability ceiling of **0.900**, and identity ARI **0.681** at match rate **0.760**. DATA

> Those Step 1 figures are **measurement limitations on the inventory**, not accuracy claims about anything in this document. **Dataset B has no ground truth**, so an unknown share of these executions is mis-bounded or mis-labelled, and nothing downstream of it can be validated against a reference. DATA

3.1 What telemetry can and cannot establish

OBSERVED	INFERRED, with support	UNOBSERVABLE
routes, origins, window titles; element identifiers; field keys; event types and counts; operator hashes; session boundaries; screenshot availability	that a process is being repeated; that a source document supplies data rather than reference; that work is duplicated across systems; that a routine is repeatable enough to automate	labour cost; true annual volume and seasonality; error and rework rates; business criticality; compliance exposure; API or import availability; production selector stability; approval obligations

INF The middle column is where this analysis does its work, and the right-hand column is why several of its conclusions are conditional. Nothing in the right-hand column is estimated anywhere in this document; each item reappears in Section 20 as a validation question.

4 The Dataset B process landscape

All figures in this section are computed from `process_inventory.json`, which is generated deterministically from the frozen Step 1 output. DATA

4.1 Inventory

Process label	n	ses	op	% time	med s	rJacc	sJacc	conc	flds	Gate
resident-tax__p02	23	12	4	19.19	56.5	0.345	0.130	0.78	28	pass
leave-apps-social-insurance__p00	27	8	4	14.74	29.4	0.604	0.326	0.89	25	pass
payroll-items__p01	24	8	4	12.77	36.9	**0.736**	0.376	**1.00**	**39**	pass
onboarding__p03	21	8	4	8.40	27.6	0.670	0.399	1.00	17	pass
leave-apps-onboarding__p06	12	4	2	7.97	38.2	0.495	0.414	0.92	11	pass
social-insurance__p04	16	11	4	7.82	29.2	0.335	0.171	0.69	23	pass
payroll-items__p05	13	5	3	6.88	41.5	**0.744**	0.268	**1.00**	8	pass
hr人事給与システム__p09	9	1	1	6.69	69.8	n/a	0.241	0.00	13	**FAIL**
onboarding__p07	11	5	3	4.70	25.5	**0.803**	0.430	1.00	13	pass
dashboard__p08	10	7	4	3.72	21.4	0.437	0.222	0.70	11	pass
leave-apps-resident-tax__p10	7	6	4	3.23	37.9	0.484	0.327	1.00	8	pass
leave-apps-payroll-items__p11	6	4	2	2.51	27.4	0.639	0.499	1.00	4	pass
onboarding-social-insurance__p12	4	1	1	1.40	22.5	0.750	0.714	1.00	12	**FAIL**

Legend. n executions · ses sessions · op operators · % time share of 7,642.7 s segmented · med s median execution seconds (relative comparison only) · rJacc mean pairwise route-set Jaccard within the label · sJacc the same over surface-shape sets · conc route concentration · flds distinct field keys. Label names abbreviated; `leave-apps` = `leave-applications`. Source: `process_inventory.json`.

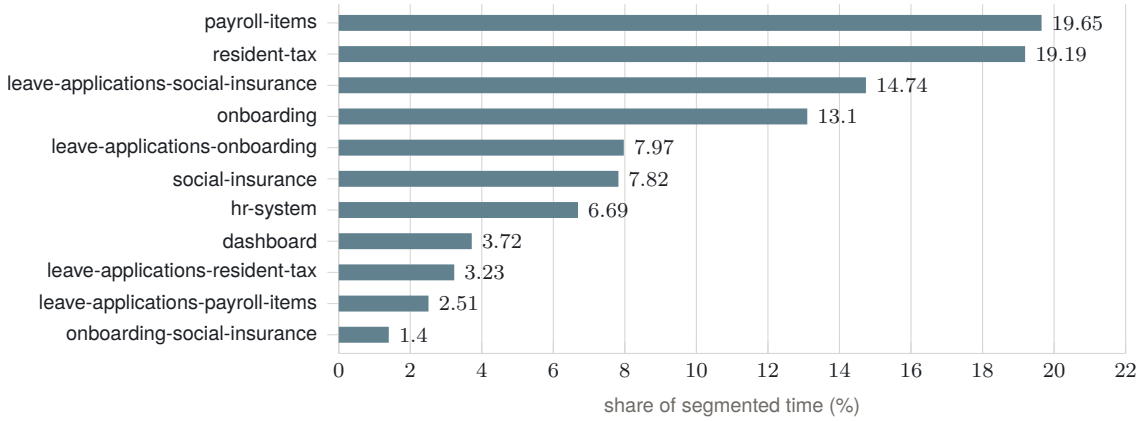


Figure 4. Share of segmented time by *route family* (sub-labels aggregated). `payroll-items` and `resident-tax` are indistinguishable on this axis alone — 19.65% and 19.19%. Durations are relative comparisons; the brief states waiting time is compressed in the capture environment. DATA

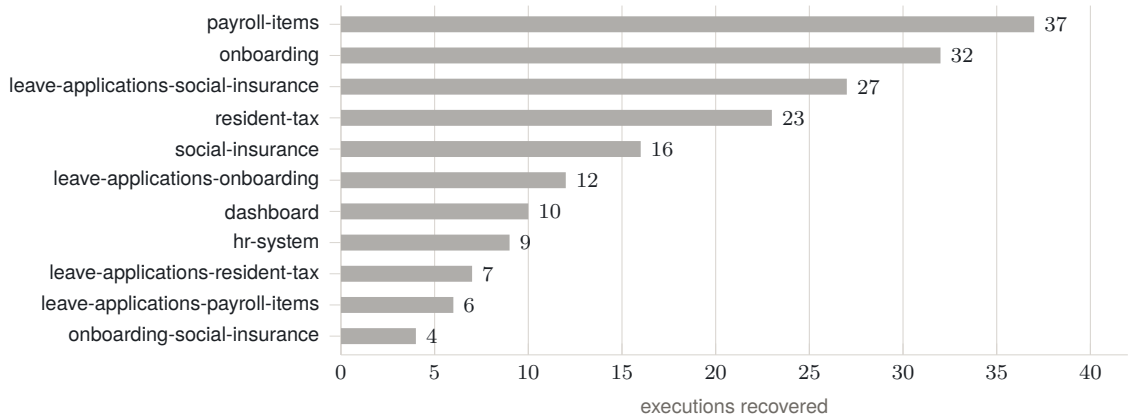


Figure 5. Executions by route family. `payroll-items` leads on count as well as time, and the ordering differs from Fig. 4 — `resident-tax` has fewer, longer executions (median 56.5 s against a dataset median nearer 30 s). DATA

5 Determinism and transfer: the suitability evidence

Suitability is not measured directly. LIT Bosco et al. define an automatable routine as one with a stable activation condition whose parameters are constant or derivable from earlier values [1]. Neither property is directly observable in desktop telemetry, so Step 2 uses two families of proxy, and says so.

Repeatability proxies. Route-set Jaccard is the mean pairwise Jaccard of the route sets of two executions of the same label; surface-set Jaccard is the same over surface shapes (`web:<origin><route>` / `app:<app>:<title>`); route concentration is the share of the label's executions whose modal route is present. **Transfer proxies.** Clipboard operations and form inputs per execution measure how much of the work is moving a value from one place to another — the operation an automation would replace.

Process label	rJacc	sJacc	identical	conc	clip/ex	form/ex	keys/ex	trans/ex	apps
`resident-tax__p02`	0.345	0.130	1	0.78	6.70	5.00	21.0	6.57	10 (5)
`leave-apps-social-insurance__p00`	0.604	0.326	1	0.89	3.30	1.93	19.2	7.11	6 (5)
`payroll-items__p01`	**0.736**	0.376	2	**1.00**	2.63	2.46	19.0	5.21	8 (4)
`onboarding__p03`	0.670	0.399	2	1.00	2.57	1.90	12.4	4.76	**2 (2)**
`leave-apps-onboarding__p06`	0.495	0.414	1	0.92	**4.75**	**3.50**	20.8	7.17	4 (3)
`social-insurance__p04`	0.335	0.171	1	0.69	2.94	2.06	10.4	6.88	7 (5)
`payroll-items__p05`	**0.744**	0.268	2	**1.00**	**4.08**	**4.00**	9.1	**1.46**	5 (4)
`onboarding__p07`	**0.803**	0.430	2	1.00	2.73	1.64	17.5	4.55	5 (2)
`dashboard__p08`	0.437	0.222	1	0.70	0.80	0.20	19.4	3.30	6 (2)
`leave-apps-resident-tax__p10`	0.484	0.327	1	1.00	3.43	3.43	7.1	1.29	2 (2)
`leave-apps-payroll-items__p11`	0.639	0.499	3	1.00	3.17	3.33	6.7	0.83	3 (3)

Legend. identical = largest number of executions sharing an *exactly* identical ordered surface shape; clip/ex, form/ex, keys/ex, trans/ex = clipboard operations, form inputs, keystrokes and application transitions per execution; apps = distinct applications, business applications in brackets. Two G1-failed labels omitted. DATA

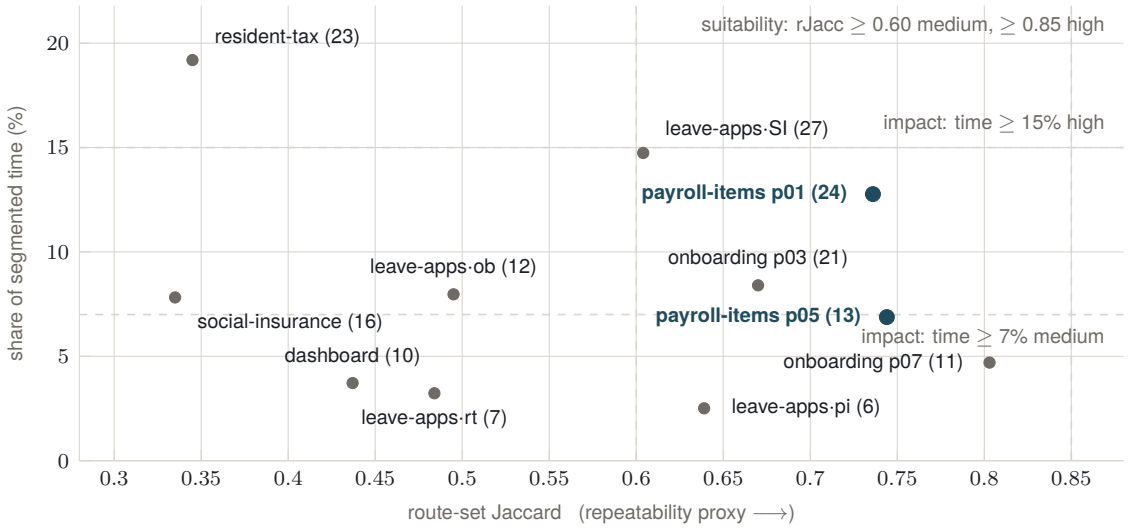


Figure 6. The tension, plotted. Executions in brackets. The upper-right region — large *and* repeatable — is empty: the largest process by time sits at the far left, and every high-repeatability label sits below the 15% impact line. `payroll-items` (marked) is the closest thing in the dataset to a compromise. Two labels that failed the evidence gate are not plotted. DATA

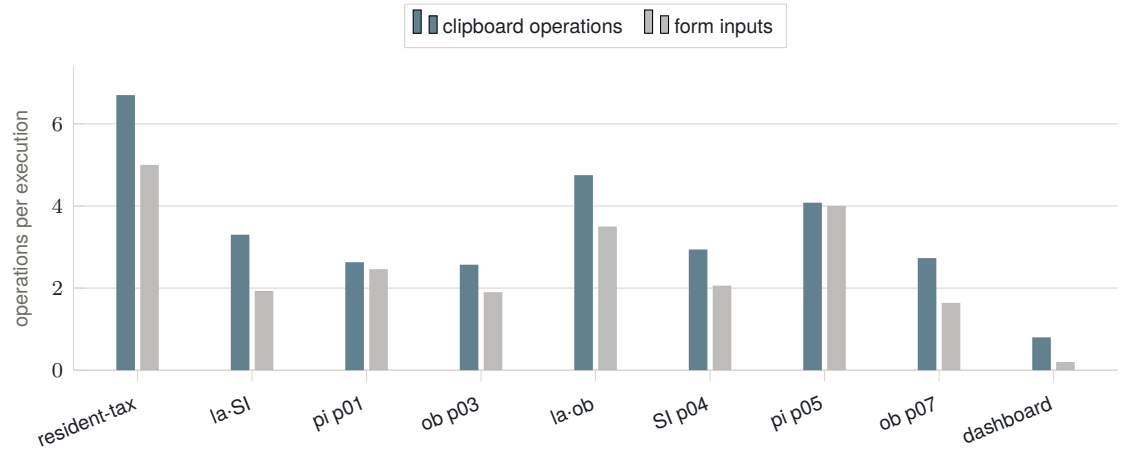


Figure 7. The transfer signature. A high bar pair means the execution consists largely of moving values between places — precisely the operation an automation replaces. `dashboard__p08` has almost none of it, which is the substantive reason to treat it as navigation overhead rather than business work. DATA

What these numbers are not. DATA Determinism here is a *proxy* for the Bosco et al. condition, not a validated production predictor. `max identical shapes` = 1 for most labels is an exact ordered-sequence test and is deliberately harsh; route-set Jaccard is the operative measure. And because Dataset B has no ground truth, every one of these values inherits Step 1's segmentation and identity error.

6 Screening: three gates, declared before the numbers

Gates are hard exclusions, declared in `analysis/step2_inventory.py` before the per-label values were inspected, and not retuned afterwards. They answer *may this label be ranked at all*, which is a different question from *how good is it*.

GATE	NAME	RULE, AS IMPLEMENTED
G1	evidence sufficiency	`n_executions >= 5 AND n_sessions >= 3 AND n_operators >= 2`
G2	is business work	`NOT (portal-root/dashboard navigation AND form_inputs/ex < 0.5 AND clipboard/ex < 1.0)`
G3	has a target surface	`distinct_routes >= 1 AND distinct_element_ids >= 1`

G1 excluded two labels DATA. hr人事給与システム__p09 — one session, one operator — also fails **G3** with zero routes and zero element identifiers: it is an application window with no web surface, so there is nothing to target even in principle. `onboarding-social-insurance__p12` — four executions, one session, one operator — fails only on evidence; it is one person's work observed once, and nothing here says it is unsuitable.

G2 excluded nothing, and the near-miss is stated rather than hidden. DATA `dashboard__p08` has `form_inputs/ex` $= 0.2$ and `clipboard/ex` $= 0.8$ — *both* below their thresholds — and survives only because its route set `[/dashboard, /payroll-items, /social-insurance]` is not *entirely* navigational. Had the rule been written on the modal route instead of all routes, it would have failed. INF The rule stands as declared; this document does not retune a gate after seeing which process it catches. Treat `dashboard__p08` as navigation overhead the gate did not catch, and note that Fig. 7 reaches the same verdict on independent evidence.

> **Why declare gates at all, rather than just ranking?** INF Because the alternative is a threshold chosen after seeing which candidate it favours. A gate that is written down first, and that visibly fails to catch something it arguably should have, is more trustworthy than a score that never had to survive that test.

7 Impact × suitability

Each label is banded **High / Medium / Low** on two ordinal axes. A band is the majority of its component bands, ties broken downward. The boundaries were fixed in advance and are printed here so a reader can disagree with them specifically rather than in general.

AXIS	COMPONENT	HIGH	MEDIUM	LOW
Impact	share of segmented time	≥ 0.15	≥ 0.07	< 0.07
	executions	≥ 20	≥ 10	< 10
	operators	≥ 3	≥ 2	< 2
	sessions	≥ 10	≥ 5	< 5
Suitability	route-set Jaccard	≥ 0.85	≥ 0.60	< 0.60
	clipboard + form inputs per execution	≥ 2.0	≥ 0.5	< 0.5
	distinct applications (*fewer is better*)	≤ 2	≤ 3	> 3
	route concentration	≥ 0.80	≥ 0.60	< 0.60

	SUITABILITY LOW	SUITABILITY MEDIUM	SUITABILITY HIGH
IMPACT HIGH	`resident-tax__p02` 19.2%, 23 executions	—	(*empty*)
IMPACT MEDIUM	`social-insurance__p04` `leave-apps-onboarding__p06` `dashboard__p08`	—	`leave-apps-social-insurance__p00` **`payroll-items__p01`** `onboarding__p03` **`payroll-items__p05`** `onboarding__p07`
IMPACT LOW	hr人事給与システム__p09 (G1/G3 fail)	`leave-apps-payroll-items__p11` `onboarding-social-insurance__p12` (G1 fail)	`leave-apps-resident-tax__p10`

The headline of the matrix. DATA The top-right cell is **empty**. The only High-impact label in Dataset B bands *Low* on suitability, and the six labels that band High on suitability are all Medium or Low on impact. **No process in Dataset B is High on both.** INF Any recommendation is therefore a trade, and this is exactly why the two axes are never summed: a composite score would have produced a single ranked number while concealing the fact that the ranking's top entry is nobody's best on either axis.

8 The shortlist

Four labels reach the shortlist: gates passed, suitability High, and enough evidence to argue about. Each is stated with the reason it makes the list, the reason it might fail, and what we would need from the client to resolve it.

CANDIDATE	exec	% time	THE ONE-LINE CASE, AND THE ONE-LINE DOUBT
`leave-apps-social-insurance__p00`	27	14.74	Most-executed label; all four operators; strongest observability (10 element ids, 25 field keys). *But* rJacc 0.604 is the floor of the High band and sJacc is 0.326 — the screens inside vary far more than the routes, so "one routine" may be two adjoining ones.
`payroll-items__p01`	24	12.77	The highest determinism of any label with 20 or more executions (rJacc 0.736), route concentration 1.00, and **39 distinct field keys** — the richest field-level evidence in the dataset. *But* 8 applications, and a measured variant split: `__p01` alone is 24 of the family's 37 executions.
`onboarding__p03`	21	8.40	The cleanest system footprint in the dataset — **2 applications**, Edge and Word — with route concentration 1.00 and all four operators. *But* 7 distinct routes at rJacc 0.670: either a genuine end-to-end flow or a Step 1 segmentation merge, and without ground truth on B we cannot tell which.
`payroll-items__p05`	13	6.88	The strongest transfer signature among suitable labels — 4.08 clipboard + 4.00 form inputs, about eight transfer operations per execution — with 1.46 transitions, the lowest of the shortlist. It sits still and copies. *But* 13 executions, 5 sessions, 3 operators: it passes G1 with margin on none of its three clauses.

INF The two `payroll-items` rows are the same business step performed two ways — `__p01` type-led (19.0 keystrokes per execution), `__p05` copy-led (9.1 keystrokes, 4.08 clipboard operations). Whether that makes them one procedure or two is a client question, and it is the single answer that would change the scope of any payroll automation by 35% of its executions.

9 The multi-system discovery

This section contains the finding that changed the interpretation of the opportunity. It came from looking inside executions at block level, not from the aggregate table.

9.1 Three origins, not one application

DATA Dataset B's browser activity resolves to **three distinct local origins**, and each carries a dominant Japanese system title:

ORIGIN	DOMINANT WINDOW TITLE	BLOCKS	DOMINANCE
`127.0.0.1:5132`	HR人事給与システム (HR / payroll)	198	153 (77%)
`127.0.0.1:5133`	財務会計システム (financial accounting)	222	183 (82%)
`127.0.0.1:5134`	受発注在庫管理システム (order / inventory)	171	138 (81%)

Blocks = all activity blocks on that origin; dominance = blocks whose window title is the dominant one. The remaining 18–23% carry a different title, which is consistent with window-title lag at the moment of a tab switch rather than with a fourth system. INF

9.2 A correction to our own earlier reading

> DATA An earlier pass of this analysis concluded from the element identifiers — twelve in the whole dataset, following one strict `btn-{module}-ok` / `{module}-note` convention — that these were *"not five systems, but one web application with a uniform widget-naming convention."*
>
> That inference was half right and is corrected here. The *widget convention* is uniform; the *origins* are three, with three different Japanese system names. INF A shared naming convention across three separately-served systems is a stronger finding than either reading alone: it means one automation shell can address all three, while the three remain genuinely separate targets that must each be written to, validated against and reconciled independently.

9.3 Route by origin

DATA Cross-tabulating route against origin at block level is what makes the shape of the work visible:

ROUTE	HR人事給与 5132	財務会計 5133	受発注在庫 5134	IN N
`/payroll-items`	44	80	33	3
`/leave-applications`	16	33	76	3
`/social-insurance`	27	9	23	3
`/onboarding`	75	31	—	2
`/resident-tax`	—	37	—	1
`/dashboard`	14	13	17	3
`/` (portal root)	22	19	22	3

Block counts. "in N" = number of origins on which the route appears with non-trivial activity. DATA

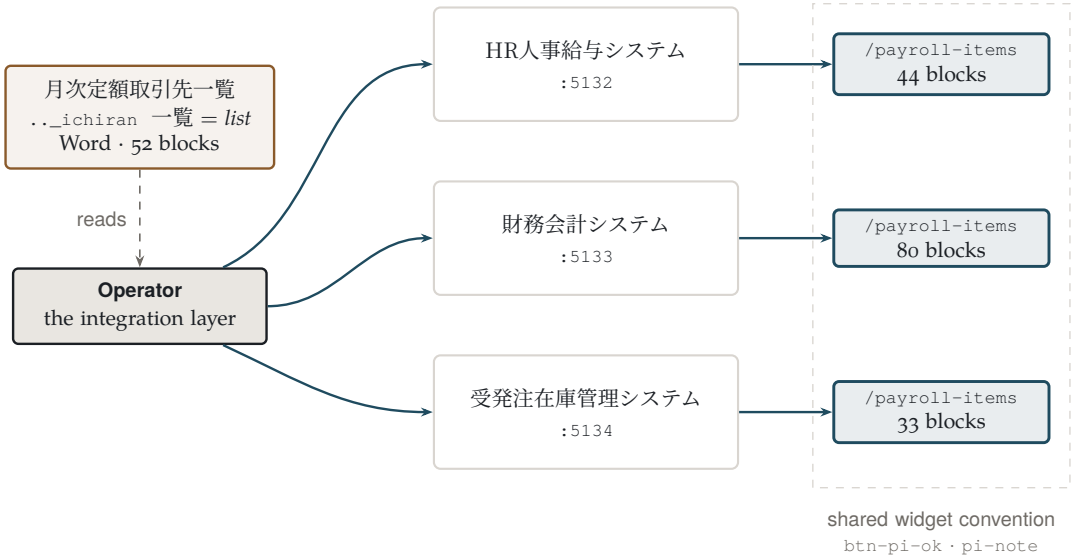


Figure 8. The observed topology. One structured source document, one human, three separately-served systems, and the same module route in all three — addressed through the same two widget identifiers. DATA The human is the only thing connecting the three systems, which is precisely the definition of the problem. INF

Why this matters for the recommendation. INF `/payroll-items` is the only module with substantial activity in all three systems. `/resident-tax` — the largest process by time — appears on *one* origin only. A candidate that touches three systems is harder to automate and more valuable to automate, because the multiplication of effort is the waste. DATA And the twelve element identifiers mean the automation surface for all three is the same two selectors.

10 The document-type discovery

The second finding came from the companion documents. Joining Word, Excel and Notepad blocks to executions by sequence overlap shows that different candidates are accompanied by different *kinds* of document, and the kind determines the solution catalogue.

CANDIDATE	DOMINANT COMPANION DOCUMENT (co-occurring blocks)	JAPANESE DOCUMENT TYPE
`payroll-items__p01`	`getsujitsu_teigaku_torihikisaki_ichiran` (33)	一覧 *list* — a table of records. **A data source.**
`leave-apps-social-insurance__p00`	`shinkui_keiyaku_tetsuzuki` (53) `shinkuitorihikisaki_touroku_tetsuzuki` (37) `keiyaku_kaijo_tetsuzuki` (33)	手続き *procedure* — how to do something. **Reference material.**
`onboarding__p03`	`nyusha_checklist_shinsotsu_batch` (36)	checklist — a control document, partly reference, partly worklist.
`resident-tax__p02`	*no single companion* — 在庫調整メモ (12), `shinkuitorihikisaki_touroku_tetsuzuki` (11), 育児休業規程 (10)	mixed メモ / 手続き / 規程. **No stable data source.**

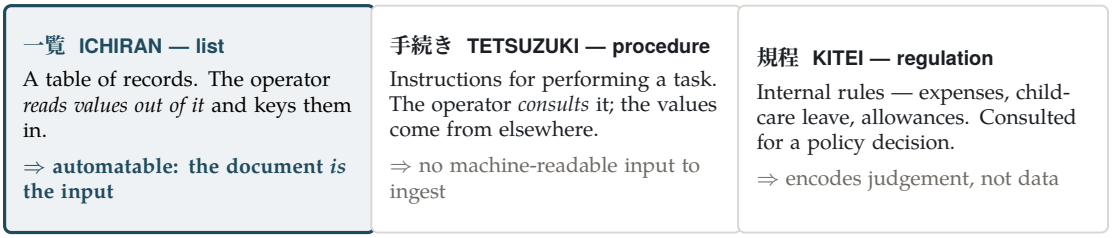


Figure 9. Three document types, three different automation implications. INF The selected candidate is the only shortlisted one whose dominant companion is an 一覧: its source of record is a structured list, which is what makes deterministic ingestion possible. The others are accompanied by material a human *reads to decide*, which is a different and harder problem. DATA INF

An honest limit on this evidence. DATA The 一覧 association is strong for `payroll-items__p01` (33 co-occurring document blocks against 8 for the next title) but **weak for `__p05`**, whose companion blocks are few and are mostly a 規程 (`kazoku_teate_kitei`, 3 blocks). INF That is consistent with `__p05` being copy-led from a source we cannot see, and it is a reason to treat the family framing as a hypothesis the client must confirm — not to quietly extend the 一覧 finding across both variants.

11 Alternatives considered, and why they were not selected

Elimination reasons are recorded per candidate, because a proposal that only argues for its winner is not auditable.

LABEL	ELIMINATED ON	REASONING
`resident-tax__p02`	suitability, *not* size	The biggest process in the dataset (19.2%, 23 executions) and among the least repeatable things in it: rJacc **0.345** — only `social-insurance__p04` is lower, at 0.335 — and sJacc **0.130**, the lowest outright; seven routes, ten applications, and no two executions sharing an identical surface shape. INF Automating it as-is would encode 23 mutually dissimilar paths. JP It is instead the clearest instance in this data of MIC's 業務の整理・見直し: a process to *review* before automating, which is a different recommendation, not a weaker one.
`social-insurance__p04`	determinism	Good spread (16 executions, 11 sessions, 4 operators) but rJacc **0.335** across 7 routes at route concentration 0.69. The same objection as `resident-tax` at half the size.
`leave-apps-onboarding__p06`	determinism and operator spread	The strongest transfer signature in the dataset (4.75 clipboard + 3.50 form inputs) and 8.0% of time — but rJacc **0.495** and only **2 operators** across 4 sessions. INF The first candidate to revisit if the client confirms it is one procedure.
`onboarding__p07`	volume, *not* quality	rJacc **0.803** is the highest in the dataset and route concentration is 1.00 — but 11 executions and 4.7% of time, and it is the smaller variant of a family whose larger variant is already shortlisted.

LABEL	ELIMINATED ON	REASONING *(continued)*
`dashboard__p08`	not business work	0.80 clipboard and 0.20 form inputs per execution (Fig. 7); it survived G2 on a technicality (§6). Navigation overhead.
`leave-apps-resident-tax__p10` `leave-apps-payroll-items__p11`	evidence, *not* unsuitability	7 and 6 executions. Both pass G1; both show high transfer signatures (3.43 and 3.17 clipboard per execution) and route concentration 1.00. INF Too thin to rank, not demonstrated to be unsuitable.
hr人事給与システム `__p09` `onboarding-social-insurance__p12`	gate G1 (and G3)	On the numbers in §6.

INF Note the shape of the reasons. Only one candidate is eliminated for being *unsuitable to automate*; the rest are eliminated for being too small, too thinly evidenced, or not business work. That distinction matters to a client, because "we lack evidence" is fixable and "this work is not repeatable" is not.

12 Reframing the opportunity: 二重入力

The selected work has a different name in every discipline, and each name carries its own solution catalogue. Choosing the wrong name imports the wrong catalogue, so the naming is done explicitly.

DISCIPLINE	TERM	FIT
Japanese enterprise IT	**二重入力** (duplicate entry) across 基幹システム (core systems)	**Closest.** Names both the problem and the setting. JP
Integration / middleware	master-data propagation; system-to-system 連携	Close; assumes a machine interface exists.
RPA	multi-system data entry	Close; assumes the UI is the interface.
Task mining	a *deterministic routine* whose parameters are constant or derivable	The formal condition our determinism proxy tested. LIT
Document automation	structured document ingestion	**Poor fit** — the source is a structured 一覧, not an unstructured scan.

The framing this document adopts

Duplicate master-data entry across siloed core systems, where the source of record is a structured document and the human is the integration layer. INF

Framing it instead as "document-to-web-form transcription" would import IDP and OCR — solutions to a problem we do not have — and would hide the multi-system dimension, which Fig. 8 shows is the actual source of the waste.

JP Japanese guidance on connecting 給与計算 (payroll) and 会計 (accounting) systems states the consequence of the disconnection plainly: where the systems are not linked, data is re-entered by hand, producing 転記ミスや科目の選択漏れ — transcription errors and missed account classifi-

cations [11]. That is a named, documented failure mode for exactly the pattern in Fig. 8, and it is the strongest external corroboration in this document that the selected work is worth addressing.

JP The same body of guidance attaches two caveats that survive *every* integration route, and both are design-relevant rather than rhetorical: **仕訳結果や計上単位の確認は 必要です** — confirmation of the posting result remains necessary — and the warning that exception scenarios, corrections and retroactive adjustments can *increase* workload if they are not designed for in advance [11, 12].

13 The automation modality landscape

"Automation" is not one technology. Six mechanisms and one non-technical option were assessed against the evidence, with no weighted score: the trade-offs are stated and the conditions under which each wins are named.

MODALITY	WINS WHEN	DETERMINISM	MAINTENANCE	FIT TO OUR EVIDENCE
API / direct integration	a documented, field-complete, permissioned API exists	high	low — versioned contract	**unknown** — the open question (§14)
Bulk / CSV import	a file interface exists and volumes are batchy	high	low–moderate; format drift	**unknown; likely to exist** JP
UI automation (selector-driven)	no machine interface is available or permitted	high *if* selectors are stable	**highest** — external UI change is the documented principal driver LIT	**workable now**; selectors recovered DATA
Workflow orchestration	multiple systems, handoffs, state across steps	n/a — a layer, not a mechanism	moderate	**needed as a layer regardless of mechanism** INF
Document processing / IDP	the source is unstructured or scanned	probabilistic	moderate	**poor fit** — source is a structured 一覧 DATA
AI-assisted / agentic	the task needs judgement or the UI is unpredictable	**low — 42.7%** on enterprise form filling LIT	unclear	**rejected for the execution path** (§17)
Process redesign	the work exists only because systems are siloed	n/a	n/a	**partly indicated — and honest to say so** JP

The conditionality is itself the design requirement

INF UI automation is the defensible default *under present uncertainty*. But because the prerequisites of the better modality are merely unverified rather than absent, a design that hard-wired UI execution would be betting on the client's systems being worse than they probably are. **The target adapter must be an interface, with UI automation as its first implementation** — so that the same definitions and validation can drive an import or API path the day one is confirmed.

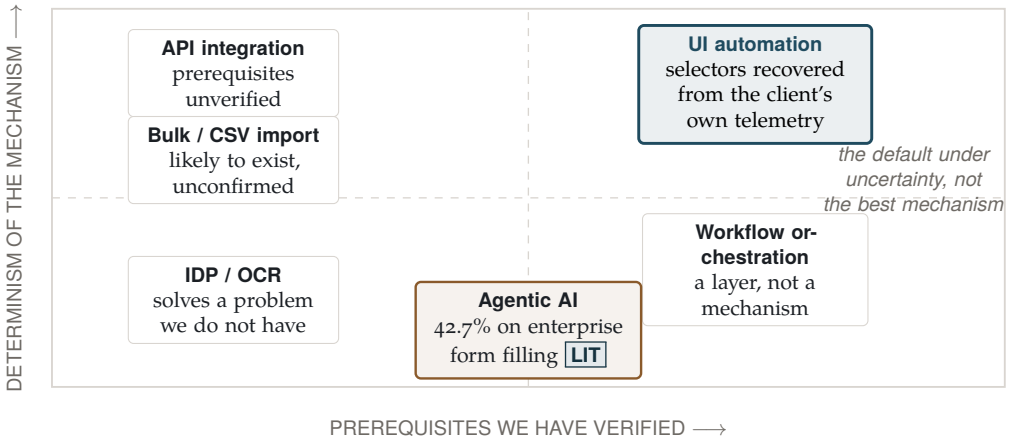


Figure 10. The design space. UI automation is selected not because it is the best mechanism, but because it is the only one whose prerequisites we have *verified*. The better mechanisms sit top-left: their prerequisites are *unverified*, not absent. INF

14 The critical question: is there an import or API path?

This is the question most likely to overturn the modality choice, so the evidence is separated into three buckets that are never mixed.

OBSERVED IN THE DATASET DATA	Three distinct origins serving the same module routes. Staff perform registration through the browser UI, alternating with a Word 一覧. Recovered widget ids follow `btn-{module}-ok` / `{module}-note`. Values are hashed (`raw_value_persisted: false`), so no payload is recoverable. **Nothing in the telemetry indicates the presence or absence of an import or API path.**
COMMON INDUSTRY CAPABILITY JP VENDOR	CSV import between payroll and accounting packages is a normal, documented capability in the Japanese market, and API integration is offered where vendors are compatible [11, 12]. It is therefore *likely* that at least one of the three systems offers a file-based route — with the stated caveat that import still requires 列の整形やアップロードの手順, column formatting and upload steps: manual work is reduced, not removed.
UNVERIFIED FOR THESE SYSTEMS	Whether *these* systems expose an import or API; whether it covers the payroll-item fields we observed; whether operators have permission to use it; and whether it is deliberately avoided for a reason we cannot see — an approval flow, an audit requirement, a data-quality gate.

Observing that staff use a browser is not evidence that no other interface exists. INF This document does not claim the observed systems lack an API. It claims we cannot see one, which is a statement about our telemetry, not about the client's architecture.

What we would need before deploying, in order. (1) Vendor documentation or an admin screen showing an import or API for the payroll-item entity in each of the three systems. (2) Field-level coverage compared against the **39 field keys** recovered DATA. (3) Operator permissions for that interface. (4) Whether an import path carries the same approval and audit obligations as UI entry. (5) If an import exists but is unused — why.

Search limitation, stated. We found no source documenting the import or API capabilities of the specific systems in Dataset B, and we did not look for one — they are a capture environment. The three-bucket

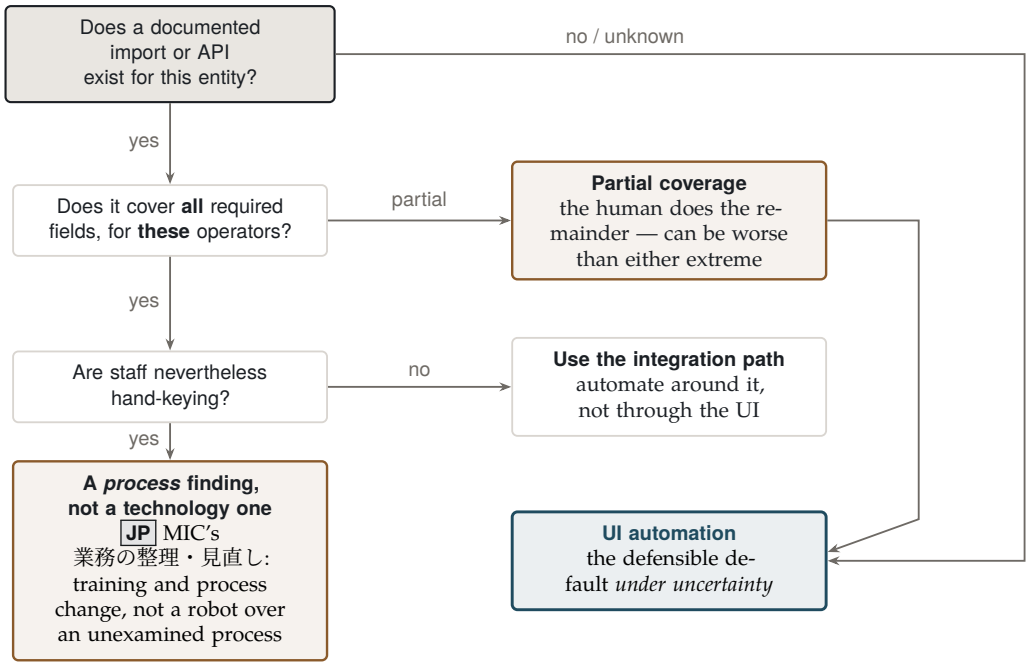


Figure 11. The decision that Step 3's architecture must not foreclose. INF The uncomfortable branch is stated deliberately: if a bulk import already exists and staff are nevertheless hand-keying, the honest recommendation is partly process change, and a robot would automate *around* an organisational problem rather than fix it.

separation above exists precisely so that absence of evidence is not recorded as evidence of absence.

15 Multi-system consistency and partial failure

Once the opportunity is understood as writing one source record into three independent systems (Fig. 8), the hardest engineering question is what happens when the second write fails after the first has succeeded.

LIT The **saga pattern** handles a sequence of writes across systems that cannot share a transaction: each step registers a *compensating transaction*, and on failure compensations run in reverse order. Because steps are retried, every operation must be **idempotent** — achieved in practice with a distinct identifier that uniquely identifies a transaction so it takes effect exactly once, now standard in payment APIs. Retry semantics are **at-least-once**, which is why idempotency is mandatory rather than optional [9].

The honest adaptation, and where we depart from the pattern

INF Sagas assume a compensating transaction *exists*. Against three enterprise UIs we do not control, it frequently will not: a payroll item, once registered and approved, may have no user-accessible reversal, and inventing one by driving a "delete" button would be a worse failure than the original. So: **prefer prevention to compensation** — validate against every target's rules before the first write; **treat each target as an independent commit** with its own state; **carry an idempotency key** per source record per target; and on partial failure **do not auto-rollback — reconcile**.

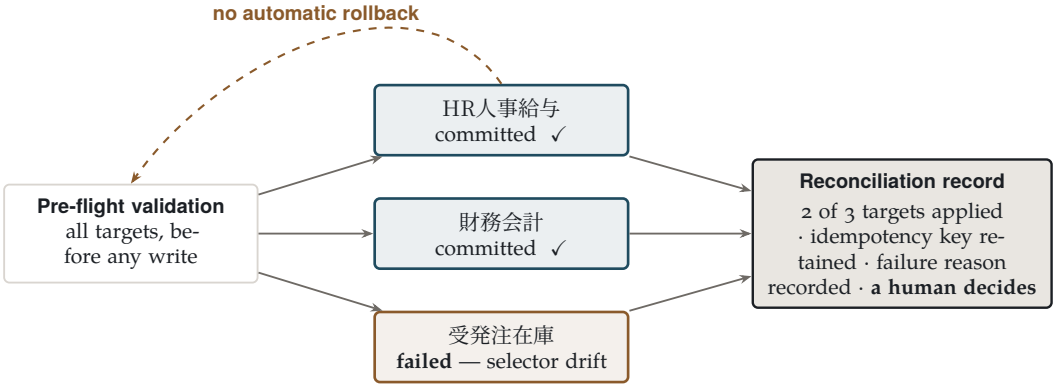


Figure 12. Partial failure, answered. INF "What if system A succeeds and system B fails?" has a defensible answer here, and it is not *undo A*. It is: A stands, B is recorded as failed with its reason, the record is flagged inconsistent, and a human decides. JP That is also what the Japanese guidance implies when it warns that corrections and retroactive adjustments increase workload if unplanned.

15.1 Failure-mode taxonomy

INF, informed by LIT and VENDOR. The column that matters is the third: it determines *where in the pipeline* each check must live.

CLASS	FAILURE	DETECT WHEN	RESPONSE
Input	malformed source, missing required field, invalid value	**before execution**	reject the record; no write attempted
Mapping	unmapped source field; ambiguous mapping	**at definition load**	fail fast — a definition error is a deploy-time bug, not a runtime one
Validation	business rule fails; target-specific rule fails	**before execution**	REQUIRES_REVIEW → human
Target	selector drift; page unavailable; timeout; unexpected UI state	at execution	stop that target, record the reason, **do not guess**
Multi-system	partial success; duplicate execution; inconsistent target states	at execution and after	reconcile; never auto-rollback
Human	rejection; correction; approval timeout	at approval	record with a change reason; corrections re-enter validation

INF Two rules follow. **Everything checkable before the first write must be checked before the first write** — because the multi-system case makes post-hoc correction expensive and sometimes impossible. And **selector drift must fail loudly**: LIT external application change is the documented principal maintenance driver [7], so the response to an unrecognised screen must be to stop and report, never to proceed on a best guess.

16 Human-in-the-loop

The question we put to the research was whether confidence scores belong here. The evidence says: mostly not.

VENDOR AWS's reference architecture for document processing with human review triggers review **when business rules fail** — "send the document ... for a human to review if any business rules fail" — rather than on a confidence threshold alone. Rules are structured objects with a *condition category* (CONFIDENCE, REQUIRED, LENGTHCHECK), a *condition type* and a *setting* [17]. Confidence is one category among several, and it exists there because extraction from a document is probabilistic.

DATA Our mapping is not probabilistic. Route Jaccard **0.74**, route concentration **1.00** and **39** stable field keys mean a source column maps to a target field deterministically. INF Attaching a confidence number to a deterministic mapping would manufacture precision of exactly the kind this analysis has refused elsewhere. **The appropriate model is a validation verdict — VALID / INVALID / REQUIRES_REVIEW — produced by declared rules.** Confidence becomes meaningful only if a later version parses a source whose structure is uncertain.

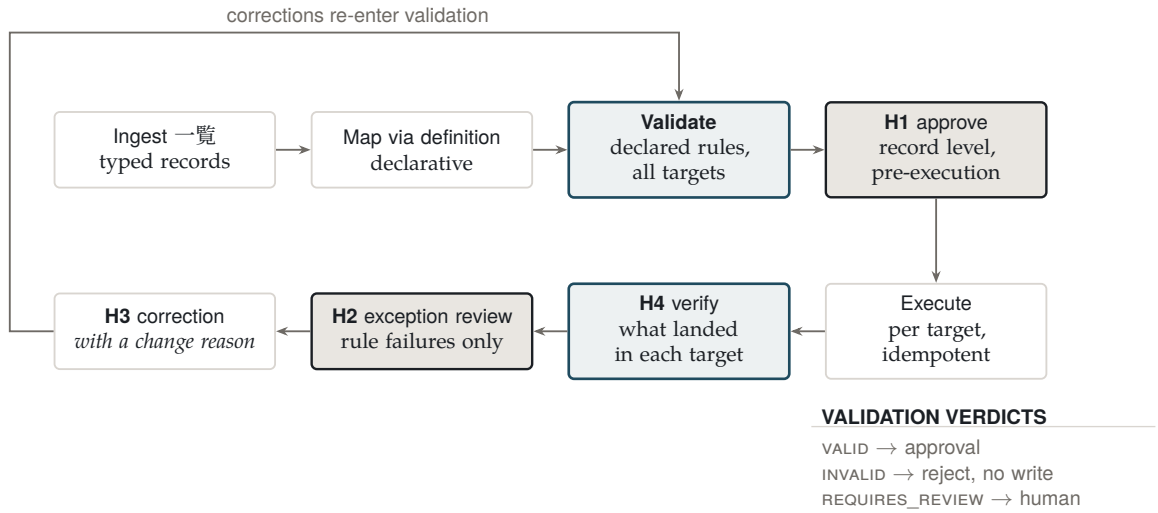


Figure 13. Where the human sits, and why there. INF Record-level approval *before* execution (the write may be irreversible, Fig. 12); exception-only review for rule failures (field-by-field review of every field would be slower than the manual process it replaces); and post-execution verification of what actually landed in each target — which JP Japanese practice says survives even full API integration.

VENDOR The same reference architecture records, for audit: reviewer identity, timestamps, time spent, worker metadata, the reviewer's input, and a required **"change reason"** free-text field [17]. INF The change reason is the part most often omitted and the part an auditor most wants.

JP And governance is not an afterthought in this market. The 野良ロボット ("stray robot") problem — automation whose owner becomes unknown and which continues to run unsupervised, in reported instances producing wrong calculations that went unnoticed — is a documented Japanese failure mode [13]. VENDOR Protiviti's guidance for RPA in SOX-relevant systems converges from the other direction: document a bot's requirements, access, design and actions at keystroke level including how it handles exceptions; avoid segregation-of-duties conflicts between whoever initiates a bot and what the bot does; and have business owners monitor transactional logs rather than letting automation run unwatched [16].

17 AI and LLM agents: assessed, and excluded from the execution path

The question was put in its strongest form: is there any part of this process where AI provides *measurable* value over deterministic automation?

> **On the execution path: no, with a number**
>
> LIT WorkArena (ServiceNow Research) benchmarks LLM agents on 33 enterprise knowledge-work tasks on a real platform, including **form filling** and service-catalogue ordering. GPT-4 with an optimised configuration reaches **42.7%** task success; the open-source state of the art reaches **17.9%**. The authors report "a considerable gap towards achieving full task automation", with failure modes including enterprise HTML exceeding 100,000 tokens and agents struggling to manipulate the UI [8].

INF Against a deterministic selector path driving a form whose element identifiers we have *recovered from the client's own telemetry*, an agent would be strictly worse on every axis that matters here — success rate, reproducibility, explainability and auditability — and its failures would be silent rather than visible. For a process with statutory exposure, that is disqualifying.

Seven candidate roles were assessed individually rather than dismissed as a class:

CANDIDATE ROLE FOR AI	VERDICT INF
Source→target **semantic field mapping**	Not needed for the observed columns — the mapping is stable and declarable once. *Possible* value as a **design-time assistant** proposing a draft mapping for a new module, with a human confirming it into a definition file. Off the execution path.
Schema inference from a new document	Same: design-time, human-confirmed, never runtime.
Anomaly detection over values	Deterministic rules (range, format, referential) cover the observed cases, and no labelled anomalies exist to validate a model against.
Exception **classification**	Plausible, but the taxonomy in §15 is small and closed. A lookup beats a model.
Natural-language process configuration	Attractive demo, no measured benefit, and it puts free text between an operator and a statutory write.
Adaptive UI recovery on selector drift	**Explicitly rejected.** The failure taxonomy requires drift to fail loudly; an AI that "figures out" a changed screen converts a visible failure into a silent wrong write.
Human-review **prioritisation**	Only meaningful at volumes we cannot observe.

Conclusion: no AI on the critical execution path. INF The one defensible role is a design-time mapping assistant whose output a human commits to a definition file — which is governance-compatible precisely because the *artefact*, not the model, is what executes. Whether even that earns its place is a Stage 3 decision.

This is the fourth time in this project that a language model has been sized against a deterministic alternative and declined on evidence. Step 1 declined it for tokenisation, for boundary decoding and for identity grouping before adopting it only for an ambiguous tail; Step 2 declines it here. The consistency is the point: the argument is about measured fitness for a specific job, not a position on AI.

18 Process definitions and reusability

The remaining architectural question is whether a reusable *definition layer* — declaring the process as data rather than coding it — is a real engineering pattern or an affectation. It is a real pattern, and the mature products converge on it.

VENDOR **UiPath's Object Repository** separates UI element definitions from process logic so that "UI elements across the project are managed, updated, and modified from a centralized place." Its taxonomy is **Applications** → **Versions** → **Screens** → **UI Elements**. A *UI Descriptor* holds the element selector, **anchor selectors** — neighbouring elements used to identify the target more robustly — screen context, captured images and metadata. The claimed benefit is that when the application changes you update descriptors once rather than editing every workflow, and that anchors let objects keep their reliability when the application receives a slightly new UI [15].

LIT The maintainability study finds that **external system changes** — vendors updating the applications bots depend on — are a principal source of maintenance burden, alongside absent development standards and poor modularity, and that updates must otherwise be applied individually across multiple bots. Among its guidelines: build reusable modules, create development standards, and perform technical reviews before bots go live [7]. It offers no quantitative maintenance figures, and we do not invent any.

DATA **Our data has exactly the shape the pattern assumes.** Three systems (Applications) sharing one widget convention, each serving the same module routes (Screens), addressed through twelve element identifiers (UI Elements). That is why the pattern transfers rather than being imported for decoration.

18.1 Open-source landscape

Only what a Stage 3 decision would actually turn on.

COMPONENT	OPTION	VERDICT
Browser automation	**Playwright** Apache-2.0 OSS	Auto-waiting removes the sleep-based flakiness that makes UI automation brittle; its selector engine supports the anchor-style robustness VENDOR describes. Preferred.
	Selenium Apache-2.0 OSS	W3C-standard and mature; weaker waiting semantics. Viable fallback, no advantage here.
Source parsing	`python-docx`, `openpyxl` MIT OSS	Adequate and boring. The source is a structured 一覧; no IDP needed.
Definition validation	`pydantic` MIT OSS	Gives definition-load-time failure — the *Mapping* class in the taxonomy — for free.
Workflow engine	Temporal / Prefect / Airflow OSS	**Reject for this scope.** They solve durable execution at scale; our state is a handful of records across three targets. LIT The maintainability warning is about *unnecessary* complexity as much as too little.
Audit storage	SQLite / JSONL OSS	Sufficient. An append-only record is what the audit requirements call for, not a database.

INF Nothing in the open-source landscape replaces the definition layer. That is ours to build, and it is the part that carries the engineering argument.

19 Design requirements derived from the research

Thirty requirements, each carrying the evidence that produced it. Stage 3 must satisfy these; *how* is Stage 3's decision. The final group cannot be demonstrated locally at all, and saying so is part of the deliverable.

ID	REQUIREMENT	EVIDENCE
FUNCTIONAL		
F1	Ingest a structured source document and produce typed records.	DATA
F2	Map source fields to target fields via a **declarative definition**, not code.	VENDOR LIT
F3	Execute one source record against **multiple target systems**, each addressed independently.	DATA
F4	Support the two observed entry variants (type-led `__p01`, copy-led `__p05`) within one definition.	DATA
F5	Add a further module by adding a definition, without engine changes.	VENDOR DATA
RELIABILITY — WHAT MUST NEVER HAPPEN		
R1	Never write without passing validation.	JP VENDOR
R2	Never write without recorded human approval.	JP VENDOR
R3	Never double-register a source record in a target — idempotency key required.	LIT
R4	Never proceed on an unrecognised screen; stop and report.	LIT
R5	Never auto-rollback a completed target write; reconcile instead.	LIT INF
R6	Never fabricate a value not present in the source.	INF
DATA		
D1	Source schema declaration.	DATA
D2	Target field keys declared per system.	DATA
D3	Element descriptors with anchors, versioned per system.	VENDOR
D4	Validation rules as data, not code.	VENDOR
D5	Idempotency key derivation rule.	LIT
HUMAN CONTROL		
H1	Record-level pre-execution approval.	JP
H2	Exception-only review for rule failures — not field by field.	VENDOR
H3	Corrections captured **with a change reason**.	VENDOR
H4	Post-execution verification of what landed in each target.	JP
H5	Separation between who initiates and what executes.	VENDOR
VALIDATION		
V1	Verdicts are VALID / INVALID / REQUIRES_REVIEW — no confidence score on a deterministic mapping.	VENDOR DATA INF
V2	Rules declared per field *and* per target.	INF
V3	All validation before the first write.	INF
AUDIT		
A1	One immutable record per execution: execution identity, definition identity *and version*, source record identity, mapped values, validation verdict and rules fired, approver and timestamp, change reason, per-target outcome and reason, target system identity, automation version, idempotency key.	VENDOR JP
A2	**Definition version** recorded with every execution.	LIT
A3	Per-target outcome and failure reason recorded separately.	LIT
INTEGRATION AND EXTENSIBILITY		
I1	The target adapter is an **interface**; UI automation is its first implementation, so an import or API path can replace it without touching definitions or validation.	JP INF

ID	REQUIREMENT	EVIDENCE (*continued*)
I2	Target system identity (origin) is a parameter, never hard-coded.	DATA
E1	New module = new definition file.	DATA
E2	New target system = new system entry, reusing the shared widget convention.	DATA
PRODUCTION PREREQUISITES — CANNOT BE DEMONSTRATED LOCALLY		
P1	Existence and field coverage of an import or API path in each system.	JP
P2	Real selector stability over time.	LIT
P3	The real validation rules of the target systems.	—
P4	Authentication, permissions and segregation of duties.	VENDOR
P5	True volumes and seasonality.	DATA
P6	Ownership and maintenance responsibility — the 野良ロボット question.	JP

20 The evidence boundary

This is the section a reviewer should read first if they intend to disagree. Everything this analysis knows, everything it inferred, and everything it refused to estimate.

KNOWN — MEASURED DATA	INFERRED — ARGUED INF	MUST BE VERIFIED WITH THE CLIENT
183 executions, 13 labels, 11 route families, 15 sessions, 4 operators, 7,642.7 s segmented.	That `payroll-items` is one business step performed two ways rather than two steps.	**P1** Does an import or API path exist, and does it cover these fields?
Per-label repeatability, transfer, route and surface statistics.	That the 一覧 is a data source and the 手続き documents are reference material.	True annual volume and seasonality — our window is roughly two hours of activity.
Three origins with three distinct Japanese system titles; `/payroll-items` present on all three.	That the three origins are three systems rather than one application behind three ports.	Error and rework rate today, and the downstream cost of a wrong record.
Twelve element identifiers on one `btn-{module}-ok` / `{module}-note` convention; 39 field keys for the selected candidate; 309 screenshots.	That UI automation is the right *default* modality, conditional on P1.	Statutory and audit obligations attaching to this process.
Companion document titles and their co-occurrence with executions.	That agentic AI is unsuitable for the execution path, sized against WorkArena. LIT	Selector stability in production, and who owns those screens.
		Who would own and maintain the automation.
		Whether `__p01` and `__p05` are one procedure or two — worth 35% of the family's executions.

Five things this document does not claim, stated so they cannot be read into it. INF

1. **No monetary figure, ROI, or FTE saving.** Labour cost is not in the telemetry, and an estimate would be invented precision.
2. **No claim that the observed systems lack an API.** We cannot see one; that is a fact about our telemetry (§14).
3. **No claim that durations are wall-clock effort.** The brief states waiting time is com-

pressed in the capture environment; durations compare processes with each other and nothing else.

4. **No claim that Step 1's Dataset A accuracy transfers.** Dataset B has no ground truth; every number here inherits an unmeasured segmentation and identity error.
5. **No claim that determinism predicts production automatability.** It is a proxy for the Bosco et al. condition [1], not a validated predictor.

21 Decision, priority order, and what would overturn it

The ordering below is a **human judgement recorded here**. It is not the output of `analysis/step2_inventory.py`, which deliberately stops at evidence, gates and ordinal bands.

#	RECOMMENDATION	REASONING
1	**`payroll-items` as a family** `__p01` + `__p05` · 37 executions · 19.6% of segmented time	Not `__p01` alone. The variants share route determinism (0.736 / 0.744) and route concentration 1.00, and differ only in *how the data gets in* — typing versus copying. Targeting the business step with both entry paths covers the family's 37 executions rather than `__p01`'s 24; the 13 additional executions are 35% of the family total, and the variant evidence is what makes that scope defensible rather than ambitious. It is also the only module present in all three systems (Fig. 8) and the only shortlisted candidate whose companion document is an 一覧 (Fig. 9).
2	`leave-applications-social-insurance__p00` 27 executions · 14.7%	The most-executed label, all four operators, strongest observability. Ranked second because its surface Jaccard of 0.326 against route Jaccard of 0.604 is the clearest signal in the dataset that the screens inside vary — and because if the client resolves the payroll variants into one procedure, candidate 1 becomes strictly better evidenced than this.
3	`onboarding__p03` 21 executions · 8.4%	The cleanest system footprint in the dataset — two applications — but its seven routes are unresolved. Ranked third *because of an open question*, not because of a measured weakness.
4	`resident-tax__p02` **redesign, not automation**	The largest process in the dataset and among the least repeatable things in it. JP MIC's position is that success requires 業務の整理・見直し rather than RPA over an unexamined process, and this is the clearest instance of it in the data. Automating it as-is would encode 23 mutually dissimilar execution paths.

What would overturn this ordering

INF It rests on repeatability and observability, with volume sizing the prize rather than setting the order. It would be overturned by: **(a)** an import or API path on the payroll module (§14), which would make UI automation the wrong *modality* — though not the wrong *process*; **(b)** a client answer that the two payroll variants are separate procedures, which would drop candidate 1 below candidate 2; or **(c)** true volumes that differ materially from this roughly two-hour window, which could reorder everything.

Modality, provisionally INF: candidate 1 shows high repeatability, deterministic-looking parameters, structured fields and no observed machine interface → *deterministic UI automation behind a swappable adapter*, conditional on P1. Candidate 2 spans two modules with a human acting as the

link → *workflow orchestration*. Candidate 4 → *redesign first*.

22 What Stage 3 must decide

This document establishes *what properties* the automation must have. It deliberately does not design it. Ten decisions are made possible — and, we think, largely determined — by the evidence above, and Stage 3 should make them explicitly rather than by default.

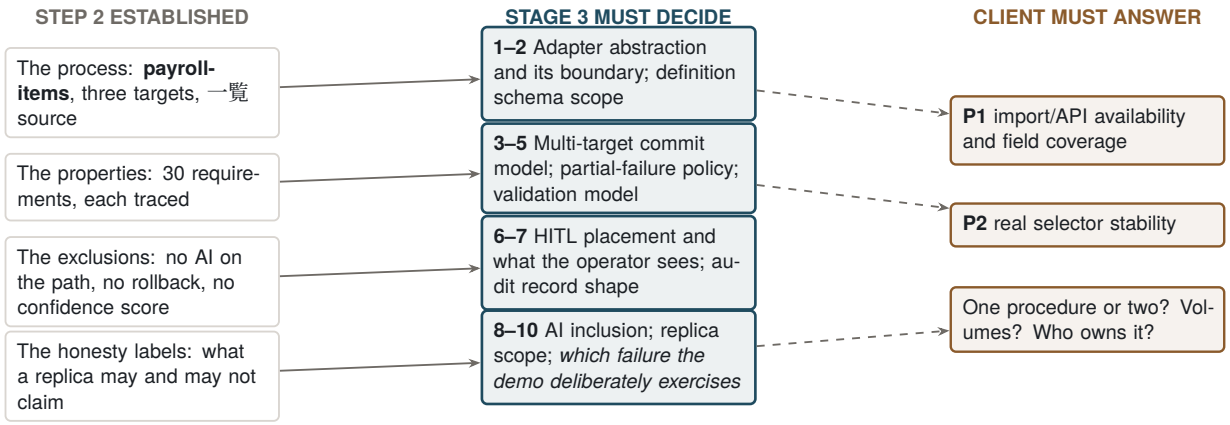


Figure 14. The bridge from Step 2 to Step 3. The dashed edges are the dependencies we cannot resolve ourselves — which is why the architecture must keep them open rather than assume an answer. INF

The ten decisions.

1. **Target adapter abstraction** — confirm the execution backend is an interface with UI automation as one implementation (I1), and fix its boundary.
2. **Definition schema scope** — which concepts are declared, and which stay in code. There is a point past which a declarative schema becomes a programming language.
3. **Multi-target execution model** — independent per-target commits with idempotency keys, versus any attempt at transactional semantics.
4. **Partial-failure policy** — reconciliation record and human decision, versus compensation (R5).
5. **Validation model** — VALID / INVALID / REQUIRES_REVIEW driven by declared rules, and explicitly *not* a confidence score (V1).
6. **HITL placement** — H1–H4, and what the operator actually sees.
7. **Audit record shape** — the A1 field list, its storage form, and its immutability guarantee.
8. **AI inclusion** — confirm exclusion from the execution path, and decide the design-time mapping assistant either in or out.
9. **Replica scope** — exactly which observed artefacts are reproduced, and how the honesty labels (TELEMETRY-RECOVERED / SYNTHETIC / REPLICA / PRODUCTION-UNVERIFIED) appear in the artefact itself.
10. **Demonstrable failure path** — which failure is deliberately exercised in the demonstration. INF A prototype that only shows the happy path evidences less than one that shows a controlled failure.

What a prototype built on this may honestly claim: *this mechanism drives the selectors the client's own telemetry contains.* **What it may not claim:** that it has run against production. INF Stating the second clearly is what makes the first believable.

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Reproducibility. Every DATA figure in this document is computed from `build/dataset_b/step2/process_inventory.json` and `build/dataset_b/step2/gates.json`, both generated deterministically from the frozen Step 1 output `build/dataset_b/segments_rich.jsonl`; the cross-tabulations in §9 and §10 are computed from `build/dataset_b/blocks.jsonl` by sequence-range overlap. No value in this document was entered by hand from memory.